Sarampreet Singh
23BCS13053
23KPG-1B

Date : ...../...../.....
Page No. : .....

Input → A = [1, 3, 5] out → 8
A = [2, 3] out → 2

Problem :- we define f(x, y) as number of different corresponding bits in the binary representation
$10^9 + 7$

Sol Approach :-

① Loop through 31 bits
② for each bit :-
Count cont1 = numbers with bit set.
count cont2 = numbers with bit unset
(N - cont1)

③ Add contribution
2 X cont1 X cont2
Q

④ Take mod

Code :->

```
#include <bits/stdc++.h>
int countbit ( int N, vector<int> & A) {
    const int MOD = 1000000007
    long long ans = 0

    for (int b = 0; b < 31; b++) {
        ~~int ones = 0~~ ;
        long long ones = 0;
```

```
for (int x : A) {
    ones += (x >> b) & 1;
}
ans = (ans + (ones * (N - ones) << 1)) %
        MOD;
}
return ans;
}

int main() {
    int N;
    cin >> N
    vector <int> v (N);
    for (int i=0; i<N; i++) cin >> v[i];
    cout << count(N, v);
    return 0;
}
```

Input = 3
~~Input~~ = [1, 3, 5]
Output ⇒ 8

Input ⇒ 2
~~Input~~ = [2, 3]
Output - 2 :